Predicting Customer Churn in Telecom Using Machine Learning

A Data Science Project

Muhammad Hannan Baig

skilledscore.com

Problem Statement

Telecom companies lose revenue due to **customer churn**.

Objective: **Predict churn** using machine learning models and **identify key drivers**.

Goal: Help the company take **proactive retention measures**.

Data Overview

Dataset: Telecom customer data

Total Records: 2000 rows x 21 Columns

Preprocessing & Handling Imbalance

- **Missing value treatment**, encoding categorical variables.
- **SMOTE** used to balance class distribution.
- **Feature scaling** applied (for Logistic Regression and XGBoost).
- Train-test split: 80-20.

Models Used

Models Used

1. Logistic Regression

- **Why Used**: Acts as a baseline model; easy to implement and interpret.
- **Strengths**: Works well with linearly separable data; good for understanding feature impact.
- **Limitations**: Struggles with complex patterns and class imbalance.

2. Random Forest Classifier

- **Why Used**: Handles non-linear relationships and performs feature selection automatically.
- **Strengths**: Robust to overfitting; good for tabular data with mixed feature types.
- **Limitations**: Less interpretable than Logistic Regression.

3. XGBoost (Extreme Gradient Boosting)

- **Why Used**: State-of-the-art performance on structured data, especially effective on imbalanced datasets.
- **Strengths**: Highly accurate; includes regularization and built-in handling of class imbalance (scale_pos_weight).
- **Limitations**: Requires careful tuning; more computationally expensive.

Hyperparameter Tuning

- Used **GridSearchCV** to optimize performance of XGBoost.
- Best results achieved after tuning max_depth, learning_rate, n_estimators, etc.

Best Performing Model

XGBoost (Tuned):

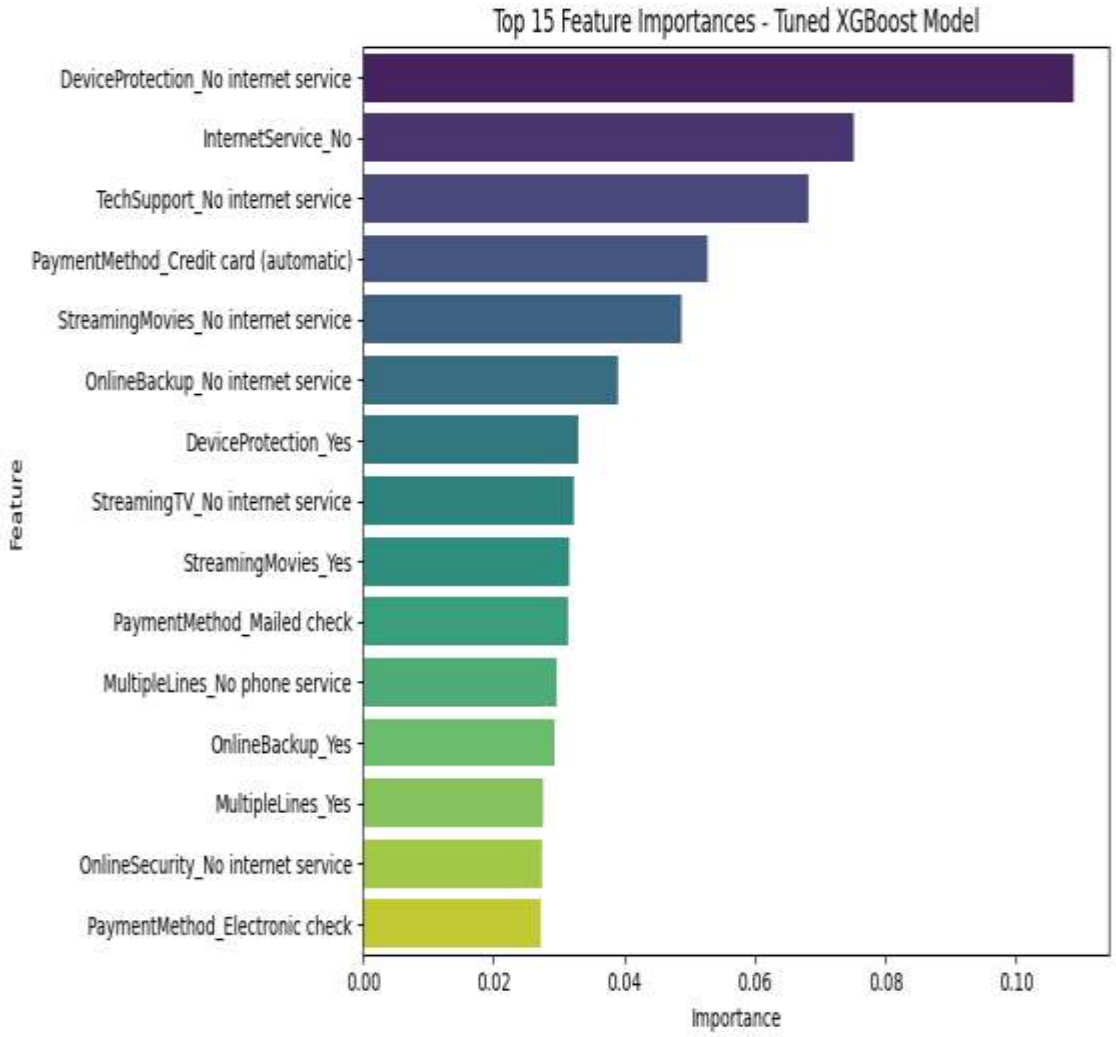
ROC-AUC: **0.56**

Recall for churn: **0.20**

Precision for churn: **0.17**

Emphasize improvement over baseline.

Feature Importance



Key drivers of churn:

- High monthly charges
- Low tenure
- No tech support, etc.

Business Insights

Focus on **customers with short tenure and high monthly charges**.

Provide **discounts or loyalty programs** to high-risk customers.

Improve **tech support and customer service** for retention.